

**ACTU 301**  
**Life Contingencies 1**

**Multiple Decrement Models**

Lecture 6

## Lecture summary

- ❖ **Multiple decrement model - expressed in terms of multiple state model**
- ❖ **Multiple Decrement Tables (MDT)**
  - several causes of decrement
  - probabilities of decrement
  - forces of decrement
- ❖ **The Associated Single Decrement Tables (ASDT)**
- ❖ **Uniform distribution of decrements**
  - in the multiple decrement context
  - in the associated single decrement context

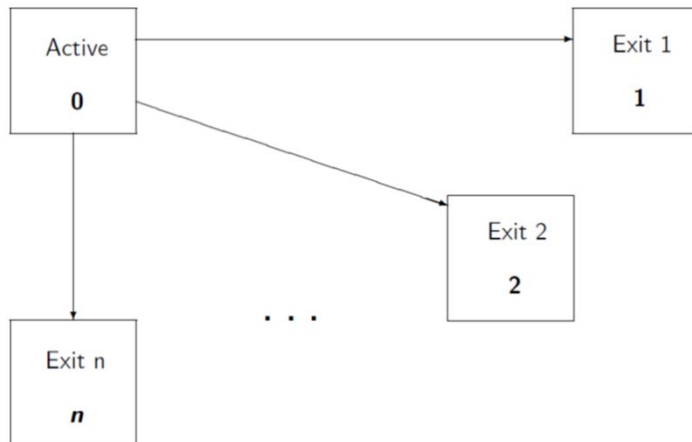
## Multiple Decrement models

A common application of multi-state model theory is in situations where a members of a population may leave due to one of several different causes. These so-called multiple decrement models can be applied to various situations.

- ❖ These models typically have one active / alive state, where each individual begins. An individual may leave the population by transitioning to any of the exit states.
- ❖ We usually only model a single transition out of the active state; we either ignore or disallow transitions among the exit states.

That is, a multiple decrement model is a multi-state model with:

- ❖ one active state
- ❖ one or more absorbing exit states



The word “**decrement**” means a reduction or lower number. Each decrement reduces the number of lives in the active state.

Suppose that we have  $n$  different causes of failure operating simultaneously on a group of lives.

These causes are often referred to as **decrements**, since they bring about a decrease in the number of lives under observation.

## Examples of multiple decrement models

Multiple decrement models are extensions of standard mortality models whereby there is simultaneous operation of several causes of decrement.

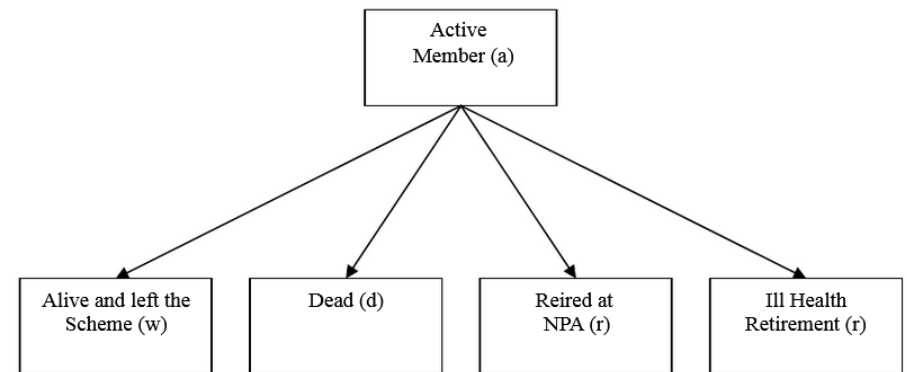
A life fails because of one of these decrements.

Examples include:

- ❖ Life insurance contract is terminated because of death/survival or withdrawal (lapse).
- ❖ An insurance contract provides coverage for disability and death, which are considered distinct claims.
- ❖ Life insurance contract pays a different benefit for different causes of death (e.g. accidental death benefits are doubled).
- ❖ Pension plan provides benefit for death, disability, employment termination and retirement.

## Simple multiple decrement model for a pension scheme

A Multiple Decrement model for a simple pension scheme with decrements for withdrawal, death, and normal and ill health retirement is as follows:



## Multiple Decrement Tables and Traditional Actuarial Notation

The probabilities for multiple decrement models are traditionally expressed in multiple decrement tables, which are analogous to the mortality tables used in our alive-dead model.

.We will number our causes (decrements) from 1 to  $m$  and use a superscript  $(i)$  to refer to cause (decrement)  $i$ .

- ❖ We will use a superscript  $(\tau)$ , to mean total.
- ❖ We begin with some cohort of  $\ell_0^{(\tau)}$  lives at age 0.
- ❖ Let  $\ell_x^{(\tau)}$  denote the number of individuals from this group who are still surviving at age  $x$ . That is, they have not succumbed to any of the  $m$  causes.
- ❖ We let  $d_x^{(i)}$  denote the number of lives leaving at age  $x$  due to decrement  $i$ . who will fail first from cause  $(i)$  between the ages of  $x$  and  $x + 1$ .

$$d_x^{(\tau)} = \sum_{j=1}^m d_x^{(j)}$$

The total number of lives who will fail from some cause between the ages of  $x$  and  $x + 1$

$$\ell_{x+1}^{(\tau)} = \ell_x^{(\tau)} - d_x^{(\tau)}$$

The number of survivors at age  $x + 1$  is equal to the number of survivors at age  $x$ , less those who failed from some cause between the ages of  $x$  and  $x + 1$

$$\ell_{x+1}^{(\tau)} = \ell_x^{(\tau)} - d_x^{(1)} - d_x^{(2)} - \dots - d_x^{(m)}$$

## Sample Double Decrement Table

Here's an excerpt of a table with two decrements:

| $x$      | $\ell_x$ | $d_x^{(1)}$ | $d_x^{(2)}$ |
|----------|----------|-------------|-------------|
| 60       | 1000     | 11          | 10          |
| 61       | 979      | 12          | 10          |
| 62       | 957      | 13          | 10          |
| 63       | 934      | 14          | 10          |
| 64       | 910      | 15          | 10          |
| $\vdots$ | $\vdots$ | $\vdots$    | $\vdots$    |

We can calculate that of the original 1000 members of this population,  $d_2^{(2)} = 10$  of them leave due to decrement 2 in the third year, and  $d_3^{(\tau)} = d_3^{(1)} + d_3^{(2)} = 24$  of them leave in the fourth year for any reason.

### Traditional Actuarial Notation: Probabilities of decrement

The probability that a life ( $x$ ) will leave the group within one year as a result of decrement  $j$ :

$$q_x^{(j)} = d_x^{(j)} / \ell_x^{(\tau)}.$$

The probability that ( $x$ ) will leave the group (regardless of decrement):

$$q_x^{(\tau)} = d_x^{(\tau)} / \ell_x^{(\tau)} = \sum_{j=1}^m d_x^{(j)} / \ell_x^{(\tau)} = \sum_{j=1}^m q_x^{(j)}.$$

The probability that ( $x$ ) will remain in the group for at least one year:

$$p_x^{(\tau)} = 1 - q_x^{(\tau)} = \ell_{x+1}^{(\tau)} / \ell_x^{(\tau)} = (\ell_x^{(\tau)} - d_x^{(\tau)}) / \ell_x^{(\tau)}.$$

We also have the probability of remaining in the group after  $n$  years

$${}_n p_x^{(\tau)} = \ell_{x+n}^{(\tau)} / \ell_x^{(\tau)} = p_x^{(\tau)} \cdot p_{x+1}^{(\tau)} \cdots p_{x+n-1}^{(\tau)}.$$

and the complement

$${}_n q_x^{(\tau)} = 1 - {}_n p_x^{(\tau)}.$$

The number of failures due to decrement  $j$  over the interval  $(x, x + n]$  is

$${}_n d_x^{(j)} = \sum_{t=0}^{n-1} d_{x+t}^{(j)}.$$

These relationships should be straightforward to follow:

$$\begin{aligned} {}_n d_x^{(j)} &= \ell_x^{(\tau)} \cdot {}_n q_x^{(j)} \\ {}_n d_x^{(\tau)} &= \ell_x^{(\tau)} \cdot {}_n q_x^{(\tau)} \end{aligned}$$

## Multiple Decrements in Continuous Models

### Total force of decrement

The total force of decrement at age  $x$  is defined as

$$\mu_x^{(\tau)} = \lim_{h \rightarrow 0} \frac{1}{h} {}_h q_x^{(\tau)} = -\frac{1}{\ell_x^{(\tau)}} \frac{d}{dx} \ell_x^{(\tau)} = -\frac{d}{dx} \log \ell_x^{(\tau)}$$

Therefore, analogous to the single decrement table, we have

$${}_t p_x^{(\tau)} = \exp - \left( \int_0^t \mu_{x+s}^{(\tau)} ds \right)$$

and

$$q_x^{(\tau)} = \int_0^1 {}_s p_x^{(\tau)} \mu_{x+s}^{(\tau)} ds$$

or, more generally

$${}_t q_x^{(\tau)} = \int_0^t {}_s p_x^{(\tau)} \mu_{x+s}^{(\tau)} ds.$$

### Force of a single decrement

The force of decrement due to decrement  $j$  is defined as:

$$\mu_x^{(j)} = -\frac{1}{\ell_x^{(\tau)}} \frac{d}{dx} \ell_x^{(j)}.$$

Notice that the denominator is NOT  $\ell_x^{(j)}$  but is rather  $\ell_x^{(\tau)}$ .

As a consequence, we see that

$$\mu_x^{(\tau)} = \sum_{j=1}^m \mu_x^{(j)}.$$

The total force of decrement is (indeed) the sum of all the other partial forces of decrement.

We can also show that

$$q_x^{(j)} = \int_0^1 {}_s p_x^{(\tau)} \mu_{x+s}^{(j)} ds.$$

### The associated single-decrement table (ASDT)

For each of the causes of decrement in an MDT, a single-decrement table can be defined showing the operation of that decrement independent of the others; called the associated single-decrement table (ASDT).

In our multiple decrement table above, the  $d_x^{(j)}$  values (and hence the calculated  $q_x^{(j)}$  values) depended on the other decrements in the table; hence,  $q_x^{(j)}$  is sometimes called a dependent probability of decrement.

If we calculate the probability of someone age  $x$  failing due to decrement  $i$  in the absence of other decrements, the resulting probability is called the independent probability of decrement, the probability of decrement in the associated single decrement table, or sometimes the absolute rate of decrement and is denoted by  $q_x'^{(i)}$ . That is, the associated probabilities in the ASDT are called absolute rates of decrements.

For example, the absolute rate of decrement due to decrement  $j$  over the interval

$$(x, x + t] \text{ is } {}_tq_x'^{(j)}$$

Note that  ${}_tq_x'^{(j)} \geq {}_tq_x^{(j)}$  for any  $x$ ;  $t$ ; and  $i$ .

Each ASDT table represents a group of lives reduced continuously by only one decrement. For example, a group subject only to death, but not to other decrements such as withdrawal.

### Formulas for Independent Probabilities in Multiple Decrement Models

The probability of  $x$  still being in the population in  $t$  years in the associated single decrement model for decrement  $i$  is denoted  ${}_tp_x'^{(i)}$ .

$${}_tp_x'^{(i)} = 1 - {}_tq_x'^{(i)}$$

$${}_tp_x^{(\tau)} = \prod_{i=1}^n {}_tp_x'^{(i)}$$

$${}_tp_x'^{(i)} = \exp \left[ - \int_0^t \mu_{x+s}^{(i)} ds \right]$$

$${}_tp_x^{(\tau)} = \exp \left[ - \int_0^t \mu_{x+s}^{(\tau)} ds \right]$$



### Link between the MDT and the ASDT

If given the absolute rates of decrements, say  $q_x^{(1)}, q_x^{(2)}, \dots, q_x^{(m)}$ , how do we derive the probabilities of decrements  $q_x^{(1)}, q_x^{(2)}, \dots, q_x^{(m)}$  in the MDT? And vice versa.

The fundamental link:  $\mu_x^{(j)} = \mu_x^{(j)}$  for all  $j = 1, 2, \dots, m$ .

Therefore, it follows that

$${}_t p_x^{(\tau)} = {}_t p_x^{(1)} \times {}_t p_x^{(2)} \times \dots \times {}_t p_x^{(m)}.$$

Furthermore, we note that

$$\begin{aligned} {}_t q_x^{(j)} &= \int_0^t {}_s p_x^{(\tau)} \cdot \mu_{x+s}^{(j)} ds \\ &= \int_0^t {}_s p_x^{(\tau)} \cdot \mu_{x+s}^{(j)} ds = \int_0^t \frac{{}_s p_x^{(\tau)}}{{}_s p_x^{(j)}} {}_s p_x^{(j)} \cdot \mu_{x+s}^{(j)} ds. \end{aligned}$$

### Fractional Age Assumptions in Multiple Decrement Models

If we have a discrete model (i.e., a multiple decrement table), we'll often need to make some assumptions regarding how the decrements act within the year:

- ❖ We need this in order to get the independent failure probabilities from the dependent ones (and vice-versa).
- ❖ We also need this additional information to do calculations for non-integral lengths of time.

### Uniform Distribution of Decrements in the Multiple-Decrement Table

One option is to assume that each decrement in the multiple decrement table is distributed uniformly over the year in the *presence* of other decrements, i.e.,  ${}_tq_x^{(j)}$  is a linear function of  $t$ .

Under this UDD assumption in the multiple decrement context:

$${}_tq_x^{(j)} = t \cdot q_x^{(j)} \quad \text{for } 0 \leq t \leq 1.$$

From this assumption, we can derive the following relationships:

$${}_tq_x^{(\tau)} = t \cdot q_x^{(\tau)} \quad {}_tp_x^{(\tau)} = 1 - t \cdot q_x^{(\tau)} \quad \mu_{x+t}^{(j)} = \frac{q_x^{(j)}}{1 - t \cdot q_x^{(\tau)}}$$

$${}_tp_x'^{(j)} = \left(1 - t \cdot q_x^{(\tau)}\right)^{q_x^{(j)}/q_x^{(\tau)}}$$

This result allows us to compute the absolute rates of decrements  $q_x'^{(j)}$  given the probabilities of decrements in the multiple decrement model. In particular, when  $t = 1$ , we have

$$q_x'^{(j)} = 1 - (1 - q_x^{(\tau)})^{q_x^{(j)}/q_x^{(\tau)}}.$$

### Uniform Distribution in the Associated Single-Decrement Tables

Another option is to assume that each decrement is distributed uniformly in its associated single decrement table, i.e., in the *absence* of other decrements, so that  ${}_tq_x'^{(j)}$  is a linear function of  $t$ .

Under this UDD assumption in the single decrement context:

$${}_tq_x'^{(j)} = t \cdot q_x'^{(j)}$$

From this assumption, we can derive various relationships. For example, for a double decrement table, we have:

$$q_x^{(1)} = q_x'^{(1)} \left(1 - \frac{1}{2} \cdot q_x'^{(2)}\right) \quad \text{and} \quad q_x^{(2)} = q_x'^{(2)} \left(1 - \frac{1}{2} \cdot q_x'^{(1)}\right)$$

Similar equations can be derived for triple decrement tables.

### Constant Force of Decrement Assumption

We could instead assume that the forces of decrement are constant within the year. This assumption results in the following formulas:

$${}_t p_x'^{(j)} = \left[ p_x'^{(j)} \right]^t \quad {}_t p_x^{(\tau)} = \left[ p_x^{(\tau)} \right]^t$$

$${}_t q_x^{(j)} = \frac{q_x^{(j)}}{q_x^{(\tau)}} \left( 1 - \left[ p_x^{(\tau)} \right]^t \right)$$

$$p_x'^{(j)} = \left[ p_x^{(\tau)} \right]^{q_x^{(j)} / q_x^{(\tau)}}$$

### Examples

### Illustration of Multiple Decrement Table

| $x$ | $\ell_x^{(\tau)}$ | $d_x^{(1)}$ | $d_x^{(2)}$ | $d_x^{(3)}$ | $q_x^{(1)}$ | $q_x^{(2)}$ | $q_x^{(3)}$ | $q_x^{(\tau)}$ | $p_x^{(\tau)}$ |
|-----|-------------------|-------------|-------------|-------------|-------------|-------------|-------------|----------------|----------------|
| 50  | 4,832,555         | 5,168       | 1,157       | 4,293       | 0.00107     | 0.00024     | 0.00089     | 0.00220        | 0.99780        |
| 51  | 4,821,927         | 5,363       | 1,206       | 5,162       | 0.00111     | 0.00025     | 0.00107     | 0.00243        | 0.99757        |
| 52  | 4,810,206         | 5,618       | 1,443       | 5,960       | 0.00117     | 0.00030     | 0.00124     | 0.00271        | 0.99729        |
| 53  | 4,797,185         | 5,929       | 1,679       | 6,840       | 0.00124     | 0.00035     | 0.00143     | 0.00301        | 0.99699        |
| 54  | 4,782,727         | 6,277       | 2,152       | 7,631       | 0.00131     | 0.00045     | 0.00160     | 0.00336        | 0.99664        |

Using the given multiple decrement table, compute and interpret the following:

$${}_2d_{51}^{(3)}$$

$${}_3p_{50}^{(\tau)}$$

$${}_2q_{53}^{(1)}$$

$${}_{2|2}q_{50}^{(2)}$$

### Example

Suppose that in a triple-decrement model, you are given constant forces of decrement, for a person now age  $x$ , as follows:

$$\begin{aligned}\mu_{x+t}^{(1)} &= b, \text{ for } t \geq 0, \\ \mu_{x+t}^{(2)} &= b, \text{ for } t \geq 0, \\ \mu_{x+t}^{(3)} &= 2b, \text{ for } t \geq 0.\end{aligned}$$

You are also given that the probability ( $x$ ) will exit the group within 3 years due to decrement 1 is 0.00884.  
Compute the length of time a person now age  $x$  is expected to remain in the triple decrement table.

**Answer (to be discussed in lecture): 83 1/3 years**

### Example

In a double decrement table where cause  $d$  is death and cause  $w$  is withdrawal, you are given:

both deaths and withdrawals are each uniformly distributed over each year of age in the double decrement table.

$$\ell_x^{(\tau)} = 1000$$

$$q_x^{(w)} = 0.48$$

$$d_x^{(d)} = 0.35d_x^{(w)}$$

Calculate  $q_x^{(d)}$  and  $q_x^{(w)}$ .

Note: One way to check your results make sense is to ensure the inequality  $q_x^{(j)} \geq q_x^{(j)}$  is satisfied.

### Example

(Decrement 1 is retirement, decrement 2 is death)

| $x$      | $\ell_x$ | $d_x^{(1)}$ | $d_x^{(2)}$ |
|----------|----------|-------------|-------------|
| 60       | 1000     | 11          | 10          |
| 61       | 979      | 12          | 10          |
| 62       | 957      | 13          | 10          |
| 63       | 934      | 14          | 10          |
| 64       | 910      | 15          | 10          |
| $\vdots$ | $\vdots$ | $\vdots$    | $\vdots$    |

**Example:** For the double decrement table above, calculate and interpret  $q'_{62}^{(1)}$  and  $q'_{62}^{(2)}$  assuming:

- (a) All retirements occur at the beginning of the year.
- (b) All retirements occur at the end of the year.
- (c) All retirements occur uniformly throughout the year.

(Assume in all cases that deaths happen throughout the year.)

**END**